Exploratory Data Analysis

Titanic Passenger Dataset

Data Analyst Internship — Task 5

Tools: Python, Pandas, Matplotlib, Seaborn

Objective

Explore passenger data using descriptive statistics and visualizations to identify distributions, relationships, missing data, and potential anomalies.

Executive summary

The commonly used Titanic dataset contains passenger demographics, ticket information, and survival outcomes. Exploratory patterns show that survival is associated with sex and passenger class. Age and fare are unevenly distributed, and missing data needs to be considered before any predictive modeling. The notebook calculates exact metrics from the CSV when run; use those run-specific outputs for any final numerical claims.

1. Dataset understanding

The notebook inspects row and column counts, data types, descriptive statistics, and category frequencies using `shape`, `info()`, `describe()`, and `value_counts()`. This provides a baseline view of the dataset before analysis.

2. Data quality

Missing values are quantified by count and percentage for each column. The analysis also checks duplicate rows and unique-value counts. Missing values should not be imputed without considering the variable, missingness pattern, and intended downstream use.

3. Distribution and anomaly analysis

Histograms are used to inspect age and fare distributions; count plots compare passenger categories; and a boxplot highlights potential high-fare outliers. Fare values are typically right-skewed, with a small number of comparatively high fares. Boxplot outliers are investigation candidates, not automatic data errors.

4. Relationships and trends

Survival rates are compared across sex and passenger class, while age distributions are compared by survival outcome. The standard Titanic data shows materially higher survival among female passengers and first-class passengers. These are associations in the observed data and should not be presented as proof of causation.

5. Multivariate analysis

A correlation heatmap summarizes linear correlations between numeric variables. A pairplot visualizes pairwise distributions and relationships for selected numeric features. Correlation alone does not establish causation, and pairwise correlation is not a complete test for multicollinearity.

Key findings and recommendations

Finding	Interpretation / next step
Survival varies by sex	Compare group survival rates rather than raw counts; investigate this feature in a predictive baseline.
Survival varies by class	Passenger class is associated with different survival outcomes; consider class as a categorical predictor.
Fare is right-skewed	Report median and interquartile range alongside the mean; investigate high-fare observations before deciding ho
Some columns have missing data	Report missingness and choose a defensible strategy for each variable before modeling.
Correlations are not causation	Use domain context and appropriate validation; avoid causal claims from EDA alone.

Conclusion

EDA reveals meaningful patterns and data-quality limitations before modeling. The next step would be to define a clear prediction target, split data appropriately, build a simple baseline, and evaluate it with suitable metrics. Numerical outputs should be copied from the executed notebook to keep the report reproducible.

Reproducibility

The accompanying notebook downloads the public Titanic CSV on first run, calculates the statistics and group survival rates directly, and generates the visualizations. Run all cells and retain the generated outputs for the final internship submission.